

4.13.79

AI25BTECH11010 - Dhanush Kumar

Question

In $\mathbb{R}^3$, let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from two planes: $P_1 : x + 2y - z + 1 = 0$, $P_2 : 2x - y + z - 1 = 0$. Let M be the locus of the feet of the perpendicular drawn from the points on L to the plane P_1 . Which of the following points lie(s) on M ?

1) $\begin{pmatrix} 0 \\ -\frac{5}{6} \\ -\frac{2}{3} \end{pmatrix}$

2) $\begin{pmatrix} -\frac{1}{6} \\ -\frac{1}{3} \\ \frac{1}{6} \end{pmatrix}$

3) $\begin{pmatrix} 0 \\ \frac{2}{3} \\ -\frac{1}{3} \end{pmatrix}$

4) $\begin{pmatrix} 0 \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}$

Solution

Step 1: Direction vector of L

$$\mathbf{r}_L(t) = t\mathbf{D}, \quad (1)$$

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad (2)$$

$$\mathbf{n}_1^T \mathbf{D} = 0, \quad \mathbf{n}_2^T \mathbf{D} = 0. \quad (3)$$

$$(\mathbf{n}_1 \quad \mathbf{n}_2)^T \mathbf{D} = 0, \quad (4)$$

$$\mathbf{D} = \begin{pmatrix} 1 \\ -3 \\ -5 \end{pmatrix}, \quad (5)$$

$$\mathbf{r}_L(t) = t\mathbf{D}. \quad (6)$$

Step 2: Foot of perpendicular to P_1

$$\mathbf{r}_M = \mathbf{r}_L(t) + \lambda \mathbf{n}_1, \quad (7)$$

$$= t\mathbf{D} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad (8)$$

$$\mathbf{n}_1^T \mathbf{r}_M + 1 = 0, \quad (9)$$

$$\mathbf{n}_1^T (t\mathbf{D} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}) + 1 = 0, \quad (10)$$

$$\mathbf{n}_1^T \mathbf{r}_L(t) = 1 \cdot t + 2(-3t) + (-1)(-5t) = 0, \quad (11)$$

$$\mathbf{n}_1^T \mathbf{n}_1 = 1^2 + 2^2 + (-1)^2 = 6, \quad (12)$$

$$6\lambda + 1 = 0 \Rightarrow \lambda = -\frac{1}{6}, \quad (13)$$

$$\mathbf{r}_M = t\mathbf{D} - \frac{1}{6} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad (14)$$

$$\mathbf{r}_M = \begin{pmatrix} t - \frac{1}{6} \\ -3t - \frac{1}{3} \\ -5t + \frac{1}{6} \end{pmatrix}. \quad (15)$$

Step 3: Checking the options

$$\text{Option (a): } \begin{pmatrix} 0 \\ -\frac{5}{6} \\ -\frac{2}{3} \end{pmatrix} = \begin{pmatrix} t - \frac{1}{6} \\ -3t - \frac{1}{3} \\ -5t + \frac{1}{6} \end{pmatrix} \Rightarrow t = \frac{1}{6}, \quad \text{lies on } M, \quad (16)$$

$$\text{Option (b): } \begin{pmatrix} -\frac{1}{6} \\ -\frac{1}{3} \\ \frac{1}{6} \end{pmatrix} = \begin{pmatrix} t - \frac{1}{6} \\ -3t - \frac{1}{3} \\ -5t + \frac{1}{6} \end{pmatrix} \Rightarrow t = 0, \quad \text{lies on } M, \quad (17)$$

$$\text{Option (c): } \begin{pmatrix} -\frac{5}{6} \\ 0 \\ \frac{2}{3} \end{pmatrix} = \begin{pmatrix} t - \frac{1}{6} \\ -3t - \frac{1}{3} \\ -5t + \frac{1}{6} \end{pmatrix} \Rightarrow t = -\frac{2}{3}, \quad y = \frac{5}{3} \neq 0, \quad \text{does not lie on } M, \quad (18)$$

$$\text{Option (d): } \begin{pmatrix} -\frac{1}{3} \\ 0 \\ \frac{2}{3} \end{pmatrix} = \begin{pmatrix} t - \frac{1}{6} \\ -3t - \frac{1}{3} \\ -5t + \frac{1}{6} \end{pmatrix} \Rightarrow t = -\frac{1}{6}, \quad y = \frac{1}{6} \neq 0, \quad \text{does not lie on } M. \quad (19)$$

Final Answer

$$\mathbf{r}_M = \begin{pmatrix} t - \frac{1}{6} \\ -3t - \frac{1}{3} \\ -5t + \frac{1}{6} \end{pmatrix}, \quad t \in \mathbb{R}, \quad (20)$$

$$\text{Points on } M = \begin{pmatrix} 0 \\ -\frac{5}{6} \\ -\frac{2}{3} \end{pmatrix}, \quad \begin{pmatrix} -\frac{1}{6} \\ -\frac{1}{3} \\ \frac{1}{6} \end{pmatrix}. \quad (21)$$